

Magnetic Effects Of Current

Questions with Answer Keys

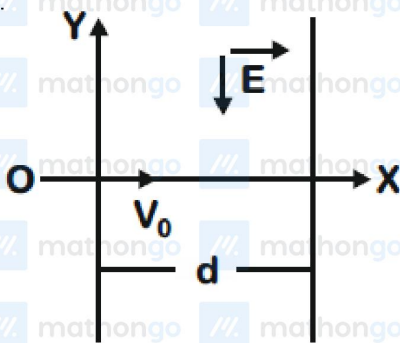
JEE Main 2020 Chapterwise

Physics

Q1 JEE Main 2020 - 2 September (Morning)

A charged particle (mass m and charge q) moves along X-axis with velocity V_0 . When it passes through the origin it enters a region

having uniform electric field $\vec{E} = -E\hat{j}$ which extends upto $x = d$. Equation of path of electron in the region $x > d$ is



(A) $y = \frac{qEd}{mV_0^2}(x - d)$

(B) $y = \frac{qEd^2}{mV_0^2}x$

(C) $y = \frac{qEd}{mV_0^2}\left(\frac{d}{2} - x\right)$

(D) $y = \frac{qEd}{mV_0^2}x$

Q2 JEE Main 2020 - 2 September (Morning)

A beam of protons with speed $4 \times 10^5 \text{ ms}^{-1}$ enters a uniform magnetic field of 0.3 T at an angle of 60° to the magnetic field. The pitch of the resulting helical path of protons is close to (Mass of the proton $= 1.67 \times 10^{-27} \text{ kg}$, charge of the proton $= 1.69 \times 10^{-19} \text{ C}$)

(A) 2cm

(B) 12cm

(C) 5cm

(D) 4cm

Q3 JEE Main 2020 - 2 September (Morning)

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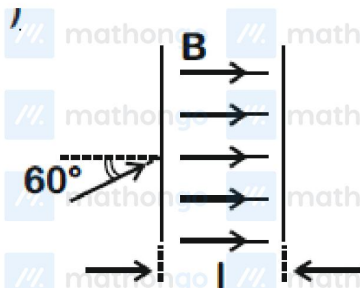
Questions with Answer Keys

Physics

A circular coil of radius 10cm is placed in a uniform magnetic field of $3.0 \times 10^{-5} \text{ T}$ with its plane perpendicular to the field initially. It is rotated at constant angular speed about an axis along the diameter of coil and perpendicular to magnetic field so that it undergoes half of rotation in 0.2 s. The maximum value of EMF induced (in μV) in the coil will be close to the integer _____.

Q4 JEE Main 2020 - 2 September (Evening)

The figure shows a region of length 'l' with a uniform magnetic field of 0.3T in it and a proton entering the region with velocity $4 \times 10^5 \text{ ms}^{-1}$ making an angle 60° with the field. If the proton completes 10 revolution by the time it cross the region shown, 'l' is close to (mass of proton = $1.67 \times 10^{-27} \text{ kg}$, charge of the proton = $1.6 \times 10^{-19} \text{ C}$)



- (A) 0.22m
- (B) 0.11m
- (C) 0.88m
- (D) 0.44m

Q5 JEE Main 2020 - 2 September (Evening)

A wire carrying current I is bent in the shape ABCDEFA as shown, where rectangle ABCDA and ADEFA are perpendicular to each other. If the sides of the rectangles are of lengths a and b, then the

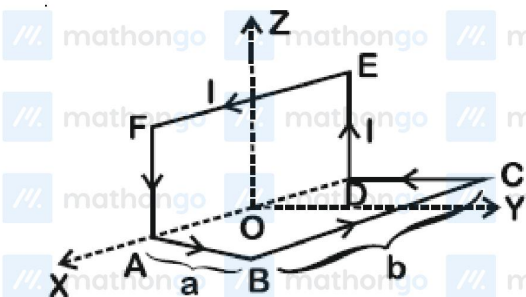
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Physics

magnitude and direction of magnetic moment of the loop ABCDEFA is



- (A) abl , along $\left(\frac{\hat{j}}{\sqrt{5}} + \frac{2\hat{k}}{\sqrt{5}}\right)$
 (B) $\sqrt{2}abl$, along $\left(\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}}\right)$
 (C) abl , along $\left(\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}}\right)$
 (D) $\sqrt{2}abl$, along $\left(\frac{\hat{j}}{\sqrt{5}} + \frac{2\hat{k}}{\sqrt{5}}\right)$

Q6 JEE Main 2020 - 3 September (Morning)

A charged particle carrying charge $1\mu\text{C}$ is moving with velocity $(2\hat{i} + 3\hat{j} + 4\hat{k})\text{ms}^{-1}$. If an external magnetic field of $(5\hat{i} + 3\hat{j} - 6\hat{k}) \times 10^{-3}\text{T}$ exists in the region where the particle is moving then the force on the particle is $\vec{F} \times 10^{-9}\text{N}$. The vector $\vec{F}$ is

- (A) $-3.0\hat{i} + 3.2\hat{j} - 0.9\hat{k}$
 (B) $-300\hat{i} + 320\hat{j} - 90\hat{k}$
 (C) $-0.30\hat{i} + 0.32\hat{j} - 0.09\hat{k}$
 (D) $-30\hat{i} + 32\hat{j} - 9\hat{k}$

Q7 JEE Main 2020 - 3 September (Morning)

Magnitude of magnetic field (in SI units) at the centre of a hexagonal shape coil of side 10cm , 50 turns and carrying current I (Ampere) in units of $\frac{\mu_0 I}{\pi}$ is

- (A) $500\sqrt{3}$

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Physics

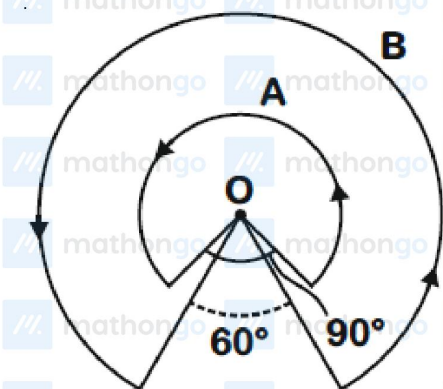
(B) $250\sqrt{3}$

(C) $50\sqrt{3}$

(D) $5\sqrt{3}$

Q8 JEE Main 2020 - 4 September (Morning)

A wire A , bent in the shape of an arc of a circle, carrying a current of 2A and having radius 2cm and another wire B , also bent in the shape of arc of a circle, carrying a current of 3A and having radius of 4cm , are placed as shown in the figure. The ratio of the magnetic fields due to the wires A and B at the common centre O is



(A) 02:05

(B) 06:05

(C) 04:06

(D) 06:04

Q9 JEE Main 2020 - 4 September (Evening)

A circular coil has moment of inertia 0.8kgm^2 around any diameter and is carrying current to produce a magnetic moment of 20Am^2 . The coil is kept initially in a vertical position and it can rotate freely around a horizontal diameter. When a uniform magnetic field of 4T is applied along the vertical, it starts rotating, around its horizontal diameter. The angular speed the coil acquires after rotating by 60° will be

(A) 10 rad s^{-1}

(B) $10\pi\text{ rad s}^{-1}$

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Physics

(C) 20 rad s^{-1}

(D) $20\pi \text{ rad s}^{-1}$

Q10 JEE Main 2020 - 5 September (Morning)

An electron is constrained to move along the y-axis with a speed of $0.1c$ (c is the speed of light) in the presence of electromagnetic wave, whose electric field is

$$\vec{E} = 30\hat{j} \sin(1.5 \times 10^7 t - 5 \times 10^{-2} x) \text{ V/m}$$

The maximum magnetic force experienced by the electron will be

(given $c = 3 \times 10^8 \text{ ms}^{-1}$ and electron charge $= 1.6 \times 10^{-19} \text{ C}$)

(A) $4.8 \times 10^{-19} \text{ N}$

(B) $2.4 \times 10^{-18} \text{ N}$

(C) $3.2 \times 10^{-18} \text{ N}$

(D) $1.6 \times 10^{-19} \text{ N}$

Q11 JEE Main 2020 - 5 September (Morning)

A square loop of side $2a$, and carrying current I , is kept in XZ plane with its centre at origin. A long wire carrying the same current I is placed parallel to the z-axis and passing through the point $(0, b, 0)$, ($b \gg a$). The magnitude of the torque on the loop about z-axis is given by

(A) $\frac{2\mu_0 I^2 a^2}{\pi b}$

(B) $\frac{\mu_0 I^2 a^2}{2\pi b}$

(C) $\frac{\mu_0 I^2 a^3}{2\pi b^2}$

(D) $\frac{2\mu_0 I^2 a^3}{\pi b^2}$

Q12 JEE Main 2020 - 6 September (Morning)

A particle of charge q and mass m is moving with a velocity $-v\hat{i}$ ($v \neq 0$) towards a large screen

placed in the $Y-Z$ plane at a distance d . If there is a magnetic field $\vec{B} = B_0 \hat{k}$, the minimum value of v for which the particle will not hit the screen is

(A) $\frac{2qdB_0}{m}$

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- (B) $\frac{qdB_0}{3m}$
(C) $\frac{qdB_0}{2m}$
(D) $\frac{qdB_0}{m}$

Q13 JEE Main 2020 - 6 September (Morning)

An electron is moving along $+\hat{x}$ direction with a velocity of $6 \times 10^6 \text{ ms}^{-1}$. It enters a region of uniform electric field of 300 V/cm pointing along $+\hat{y}$ direction. The magnitude and direction of the magnetic field set up in this region such that the electron keeps moving along the x direction will be

- (A) $5 \times 10^{-3} \text{ T}$, along $+\hat{z}$ direction
(B) $5 \times 10^{-3} \text{ T}$, along $-\hat{z}$ direction
(C) $3 \times 10^{-4} \text{ T}$, along $+\hat{z}$ direction
(D) $3 \times 10^{-4} \text{ T}$, along $-\hat{z}$ direction

Q14 JEE Main 2020 - 6 September (Evening)

A square loop of side $2a$ and carrying current I is kept in xz plane with its centre at origin. A long wire carrying the same current I is placed parallel to z -axis and passing through point $(0, b, 0)$, ($b \gg a$). The magnitude of torque on the loop about z -axis will be

- (A) $\frac{\mu_0 I^2 a^2}{2\pi b}$
(B) $\frac{2\mu_0 I^2 a^2 b}{\pi(a^2 + b^2)}$
(C) $\frac{2\pi(a^2 + b^2)}{\mu_0 I^2 a^2 b}$
(D) $\frac{2\mu_0 I^2 a^2}{\pi b}$

Q15 JEE Main 2020 - 6 September (Evening)

A charged particle going around in a circle can be considered to be a current loop. A particle of mass m carrying charge q is moving in a plane with speed v under the influence of magnetic field $\vec{B}$. The magnetic moment of this moving particle :

- (A) $-\frac{mv^2 \vec{B}}{2B^2}$

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Physics

(B) $\frac{mv^2\vec{B}}{2B^2}$

(C) $-\frac{mv^2\vec{B}}{2\pi B^2}$

(D) $-\frac{mv^2\vec{B}}{B^2}$

Q16 JEE Main 2020 - 7 January (Morning)

If the magnetic field in a plane electromagnetic wave is given by

$\vec{B} = 3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{j} T$, then what will be expression for electric field?

(A) $\vec{E} = 3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{j} V/m$

(B) $\vec{E} = 3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{i} V/m$

(C) $\vec{E} = 60 \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} V/m$

(D) $\vec{E} = 9 \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} V/m$

Q17 JEE Main 2020 - 7 January (Morning)

Consider a circular coil of wire carrying constant current I , forming a magnetic dipole. The magnetic flux through an infinite plane that contains the circular coil and excluding the circular coil area is given by ϕ_i . The magnetic flux through the area of the circular coil area is given by ϕ_o . Which of the following option is correct?

(A) $\phi_i > \phi_o$

(B) $\phi_i < \phi_o$

(C) $\phi_i = -\phi_o$

(D) $\phi_i = \phi_o$

Q18 JEE Main 2020 - 7 January (Morning)

A loop $ABCDEF A$ of straight edges has six corner points $A(0, 0, 0)$, $B(5, 0, 0)$, $C(5, 5, 0)$,

$D(0, 5, 0)$, $E(0, 5, 5)$ and $F(0, 0, 5)$. The magnetic field in this region is $\vec{B} = (3\hat{i} + 4\hat{k}) T$. The quantity of flux through the loop $ABCDEF A$ (in Wb) is

Q19 JEE Main 2020 - 7 January (Evening)

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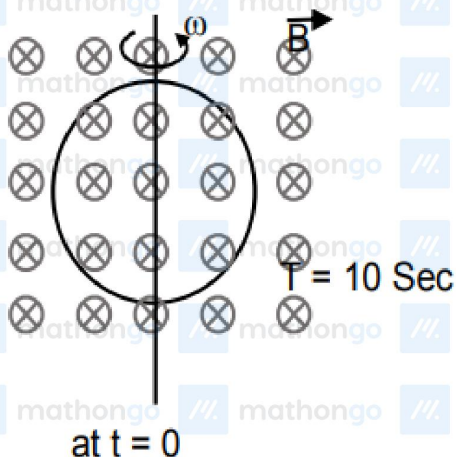
Physics

A particle of mass m and charge q has an initial velocity $\vec{v} = v_0 \hat{j}$. If an electric field $\vec{E} = E_0 \hat{i}$ and magnetic field $\vec{B} = B_0 \hat{i}$ act on the particle, its speed will double after a time :

- (A) $t = \frac{mv_0 \sqrt{3}}{qE}$
- (B) $t = \frac{mv_0 \sqrt{2}}{qE}$
- (C) $t = \frac{mv_0}{qE}$
- (D) $t = \frac{mv_0}{2qE}$

Q20 JEE Main 2020 - 7 January (Evening)

A planar loop of wire rotates in a uniform magnetic field. Initially at $t = 0$, the plane of the loop is perpendicular to the magnetic field. If it rotates with a period of 10s about an axis in its plane then the magnitude of induced emf will be maximum and minimum, respectively at :



- (A) 2.5 sec, 5 sec
- (B) 5 sec, 7.5 sec
- (C) 2.5 sec, 7.5 sec
- (D) 10 sec, 5 sec

Q21 JEE Main 2020 - 7 January (Evening)

The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0 \frac{\hat{i} + \hat{j}}{\sqrt{2}} \cos(kz + \omega t)$. At $t = 0$, a positively charged particle is at the point $(x, y, z) = (0, 0, \frac{\pi}{k})$. If its instantaneous velocity at $(t = 0)$

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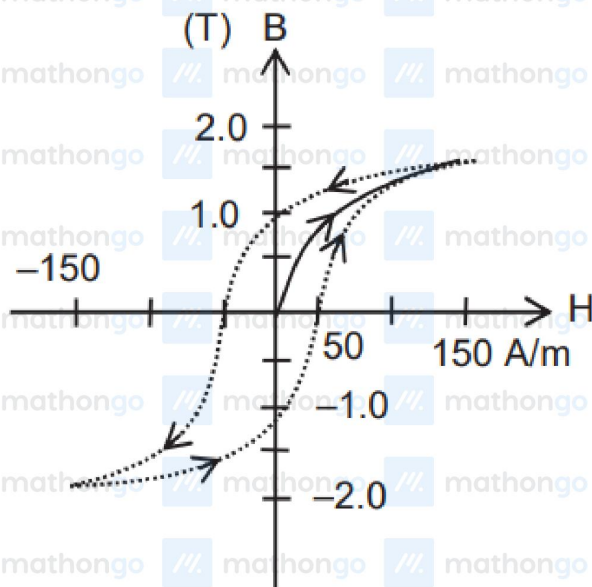
Physics

is $v_o \hat{k}$, the force acting on it due to the wave is :

- (A) $\hat{f} = 0$
- (B) Antiparallel to $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$
- (C) Parallel to $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$
- (D) $\hat{k}$

Q22 JEE Main 2020 - 7 January (Evening)

The figure gives experimentally measured B vs. H variation in a ferromagnetic material. The retentivity, co-ercivity and saturation, respectively, of the material are :



- (A) 50 A/m , 1 T , 1.5 T
- (B) 1.5 T , 50 A/m , 1 T
- (C) 1 T , 50 A/m , 1.5 T
- (D) 50 A/m , 1.5 T , 1 T

Q23 JEE Main 2020 - 8 January (Morning)

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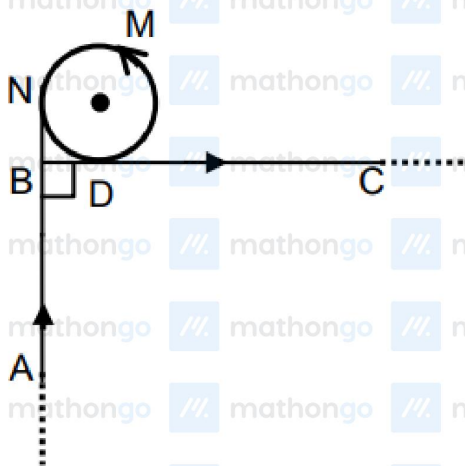
Physics

Proton with kinetic energy of 1MeV moves from south to north. It gets an acceleration of 10^{12}m/s^2 by an applied magnetic field (west to east). The value of magnetic field: (Rest mass of proton is $1.6 \times 10^{-27}\text{Kg}$)

- (A) 0.71 mT
- (B) 7.1 mT
- (C) 71 mT
- (D) 710 mT

Q24 JEE Main 2020 - 8 January (Evening)

A very long wire $ABDMNDC$ is shown in figure carrying current i . AB and BC parts are straight, long and at right angle. At D wire forms a circular turn $DMND$ of radius R . AB , BC parts are tangential to circular turn at N and D . Magnetic field at the center of circle is :



- (A) $\frac{\mu_0 i}{2\pi R} \left[\frac{-1}{\sqrt{2}} + \pi \right]$
- (B) $\frac{\mu_0 i}{2\pi R} [\pi - 1]$
- (C) $\frac{\mu_0 i}{2R}$
- (D) $\frac{\mu_0 i}{2\pi R} \left[\frac{1}{\sqrt{2}} + \pi \right]$

Q25 JEE Main 2020 - 8 January (Evening)

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Questions with Answer Keys

Physics

A plane electromagnetic wave of frequency 25GHz is propagating in vacuum along the z -direction. At a particular point in space and time, the magnetic field is given by $\vec{B} = 5 \times 10^{-8} \hat{j} \text{T}$. The corresponding electric field $\vec{E}$ is (Speed of light $c = 3 \times 10^8 \text{ms}^{-1}$)

- (A) $1.66 \times 10^{-18} \hat{i} \text{V/m}$
- (B) $15 \hat{i} \text{V/m}$
- (C) $-15 \hat{i} \text{V/m}$
- (D) $18 \hat{i} \text{V/m}$

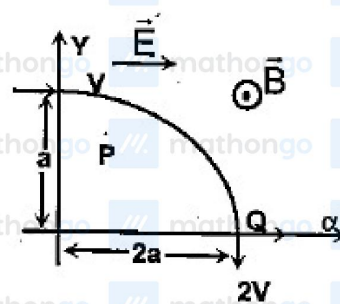
Q26 JEE Main 2020 - 9 January (Morning)

A long, straight wire of radius a carries a current distributed uniformly over its cross-section. The ratio of the magnetic fields due to the wire at distance $\frac{a}{3}$ and $2a$, respectively from the axis of the wire is :

- (A) $\frac{3}{5}$
- (B) $\frac{2}{3}$
- (C) $\frac{1}{2}$
- (D) $\frac{4}{3}$

Q27 JEE Main 2020 - 9 January (Morning)

A charged particle of mass m and charge q moving under the influence of uniform electric field $E\hat{i}$ and a uniform magnetic field $B\hat{k}$ follows a trajectory from point P to Q as shown in figure. The velocities at P and Q are respectively, $v\hat{i}$ and $-2v\hat{j}$. Then which of the following statements (A, B, C, D) are the



correct ? (Trajectory shown is schematic and not to scale)

- (A) Magnitude of electric field $E_0 = \frac{3}{4} \left(\frac{mv^2}{qa} \right)$

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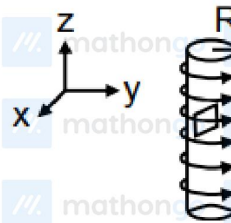
Questions with Answer Keys

Physics

- (B) Rate of work done at point A is $\frac{3}{4} \frac{mv^3}{a}$
- (C) Rate of work done by both fields at point B is zero.
- (D) Change in angular momentum w.r.t. origin is $2 maV$

Q28 JEE Main 2020 - 9 January (Evening)

An electron gun is placed inside a long solenoid of radius R on its axis. The solenoid has n turns/length and carries a current I . The electron gun shoots an electron along the radius of the solenoid with speed v . If the electron does not hit the surface of the solenoid, maximum possible value



of v is (all symbols have their standard meaning)

- (A) $\frac{e\mu_0 nIR}{2m}$
- (B) $\frac{e\mu_0 nIR}{m}$
- (C) $\frac{m}{2e\mu_0 nIR}$
- (D) $\frac{m}{e\mu_0 nIR}$

Q29 JEE Main 2020 - 9 January (Evening)

A small circular loop of conducting wire has radius a and carries current I . It is placed in a uniform magnetic field B perpendicular to its plane such that when rotated slightly about its diameter and released, it starts performing simple harmonic motion of time period T . If the mass of the loop is m then

- (A) $T = \sqrt{\frac{\pi m}{IB}}$
- (B) $T = \sqrt{\frac{2m}{IB}}$
- (C) $T = \sqrt{\frac{\pi m}{2IB}}$
- (D) $T = \sqrt{\frac{2\pi m}{IB}}$

Magnetic Effects Of Current

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Physics

Q30 JEE Main 2020 - 9 January (Evening)

A plane electromagnetic wave is propagating along the direction $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$ with its polarization along the direction $\hat{k}$. The correct form of the magnetic field of the wave would be (here B_0 is an appropriate constant) :

(A) $B_0 \frac{\hat{i} + \hat{j}}{\sqrt{2}} \cos\left(\omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$

(B) $B_0 \frac{\hat{i} - \hat{j}}{\sqrt{2}} \cos\left(\omega t + k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$

(C) $B_0 \frac{\hat{i} - \hat{j}}{\sqrt{2}} \cos\left(\omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$

(D) $B_0 \hat{k} \cos\left(\omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$

Magnetic Effects Of Current

Questions with Answer Keys

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Physics

Answer Key

Q1 (C)

Q5 (B)

Q9 (None of the option)

Q13 (A)

Q17 (C)

Q21 (B)

Q25 (B)

Q29 (D)

Q2 (D)

Q6 (D)

Q10 (A)

Q14 (B)

Q18 (175)

Q22 (C)

Q26 (B)

Q30 (C)

Q3 (15)

Q7 (A)

Q11 (A)

Q15 (A)

Q19 (A)

Q23 (A)

Q27 (A)

Q4 (D)

Q8 (B)

Q12 (D)

Q16 (D)

Q20 (A)

Q24 (D)

Q28 (A)

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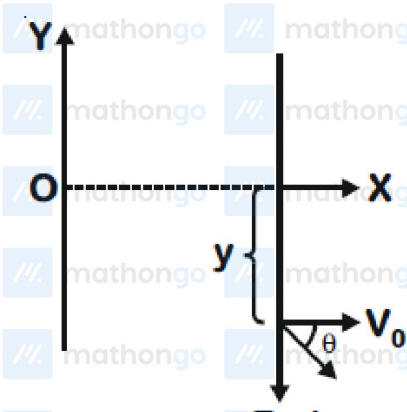
Magnetic Effects Of Current

Hints & Solutions

JEE Main 2020 Chapterwise

Physics

Q1



$$\frac{qE}{m} \frac{d}{V_0}$$

$$y = \frac{1}{2} \left(\frac{qE}{m} \right) \left(\frac{d}{V_0} \right)^2$$

$$y = -\frac{qEd}{mV_0^2} x + C$$

$$\text{At } x = d, y = -\frac{qEd^2}{2mV_0^2}$$

$$\Rightarrow C = \frac{qEd^2}{2mV_0^2}$$

$$y = -\frac{qEd}{mV_0^2} x + \frac{qEd^2}{2mV_0^2}$$

$$y = \frac{qEd}{mV_0^2} \left(\frac{d}{2} - x \right)$$

Q2

$$\text{Pitch} = v \cos \theta \times \frac{2\pi m}{qB}$$

$$= 4 \times 10^5 \times \frac{1}{2} \times \frac{2\pi \times 1.67 \times 10^{-27}}{1.69 \times 10^{-19} \times 0.3}$$

$$= 4 \text{ cm}$$

Q3

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$$\omega = \frac{\pi \times 10}{2} = 5\pi \text{ rad/s}$$

$$\phi = \pi r^2 \cdot B \cos \omega t$$

$$\therefore |\epsilon| = \left(\frac{d\phi}{dt} \right) = \pi r^2 B \omega \sin \omega t$$

$$\therefore |\epsilon|_{\max} = \pi r^2 B \omega$$

$$= \frac{\pi \times 1}{100} \times 3 \times 10^{-5} \times 5 \times \pi$$

$$\text{Taking } \pi^2 \approx 10 \text{ we get } (\epsilon_{\max}) = 15 \times 10^{-6} \text{ volt}$$

Q4

$$(v_0 \cos \theta) \times \Delta T = 1$$

$$\text{and, } \Delta T = 10 \times \frac{2\pi m}{qB}$$

$$\therefore 4 \times 10^5 \times \frac{1}{2} \times \frac{10 \times 2\pi \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.3} = 1$$

$$\Rightarrow 1 = 0.44 \text{ m}$$

Q5

$$\vec{M} = \vec{M}_1 + \vec{M}_2$$

$$\vec{M}_1 = abI \hat{j}$$

$$\vec{M}_2 = abI \hat{k}$$

$$\vec{M} = \sqrt{2}abI \left(\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}} \right)$$

Q6

Magnetic Effects Of Current

Hints & Solutions

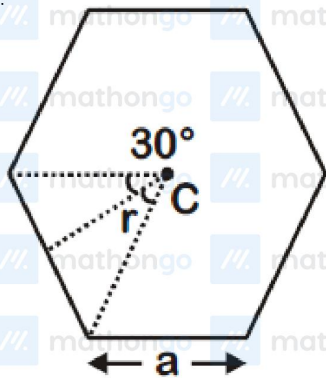
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Physics

$$\begin{aligned} \vec{F} &= q(\vec{v} \times \vec{B}) \\ &= (1 \times 10^{-6}) [2\hat{i} + 3\hat{j} + 4\hat{k}] \times [5\hat{i} + 3\hat{j} - 6\hat{k}] \times 10^{-3} \\ &= (-30\hat{i} + 32\hat{j} - 9\hat{k})10^{-9}\text{N} \end{aligned}$$

Q7

$$B_C = N \cdot \left(6 \cdot \frac{\mu_0 i}{4\pi r} (\sin 30^\circ + \sin 30^\circ) \right)$$



$$\begin{aligned} r &= \frac{\sqrt{3}}{2} a \\ &= \frac{300}{4r} \frac{\mu_0 i}{\pi} \\ &= 500\sqrt{3} \frac{\mu_0 i}{\pi} \end{aligned}$$

Q8

$$B_A = \frac{(\mu_0)(2)}{(2)(2)} \frac{3(\pi/2)}{2\pi} \dots (1)$$

$$B_B = \frac{(\mu_0)(3)}{(2)(4)} \frac{5\pi/3}{2\pi} \dots (2)$$

From (1) and (2)

$$\frac{B_A}{B_B} = \frac{6}{5}$$

Q9

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Physics

$$|U_f - U_i| = \frac{1}{2} I \omega^2$$

$$20 \times 4 \times \frac{\sqrt{3}}{2} - 0 = \frac{1}{2} \times (0.8) \omega^2$$

$$\Rightarrow \omega = 13.16 \text{ rad/s}$$

Q10

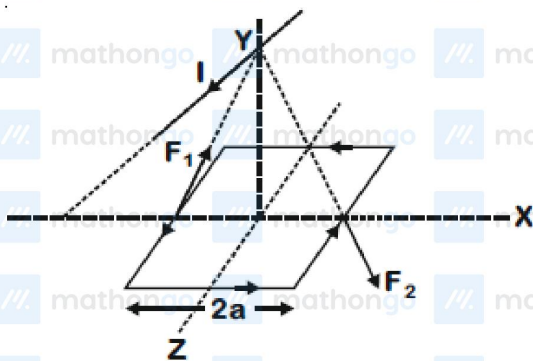
$$F_m^{\max} = (q)VB_0$$

$$\frac{E_0}{B_0} = c \Rightarrow B_0 = \left(\frac{E_0}{c} \right)$$

$$\Rightarrow F_m^{\max} = (1.6 \times 10^{-19}) (0.1c) \left(\frac{E_0}{c} \right)$$

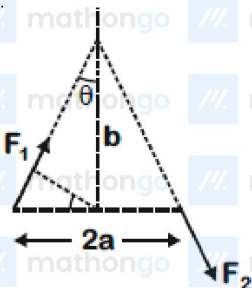
$$\simeq 4.8 \times 10^{-19} \text{ N}$$

Q11



$$F_1 = \frac{\mu_0 I^2}{2\pi r} 2a \text{ toward wire}$$

$$F_2 = \frac{\mu_0 I^2}{2\pi r} 2a \text{ away from wire}$$



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Physics

Torque about z-axis

$$= F_1 \cdot a \cos \theta + F_2 \cdot a \cos \theta$$

$$= 2 \left(\frac{\mu_0^2 2a}{2\pi r} \right) \cdot \frac{b}{r}$$

$$= \frac{2\mu_0 I^2 a^2 b}{\pi(a^2 + b^2)}$$

$$= \frac{2\mu_0 I^2 a^2}{\pi b} \quad \text{for } (b \gg a)$$

Q12

$$r = \frac{mv}{qB_0}$$

To not collide, $r < d$

$$\Rightarrow \frac{mv}{qB_0} < d$$

$$\therefore v_{\max} = \frac{qB_0 d}{m}$$

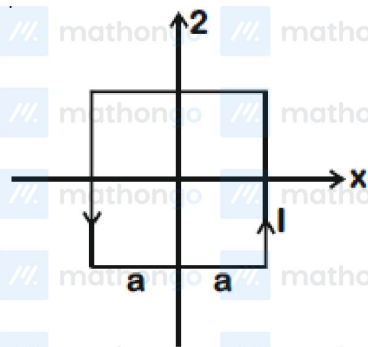
Note : It should be maximum instead of minimum.

Q13

$$\vec{F} = q(\vec{E} + \vec{V} \times \vec{B})$$

$$\vec{E} + \vec{V} \times \vec{B} = 0$$

Q14



$$\tau = (\mathbf{f}_{\perp}) \mathbf{r}$$

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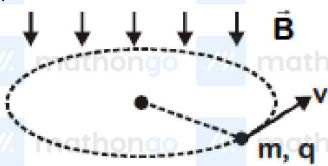
Physics

$$f_{\perp} = (1) \frac{\mu \cdot 1}{2\pi\sqrt{a^2+b^2}} \frac{(b)2a}{\sqrt{a^2+b^2}}$$

$$\tau_{\text{total}} = \frac{2\mu_0 I^2 a^2 b}{\pi(a^2+b^2)}$$

Q15

$$\vec{M} = \frac{1}{2} q \vee R(-\hat{B})$$



$$= \frac{qv}{2} \left(\frac{mv}{qB} \right) \left(\frac{-\vec{B}}{B} \right)$$

$$= -\frac{mv^2}{2B^2} \vec{B}$$

Q16

$$\frac{E_0}{B_0} = C \text{ (speed of light in vacuum)}$$

$$E_0 = B_0 C = 3 \times 10^{-8} \times 3 \times 10^8$$

$$= 9 \text{ N/C}$$

$$\text{So } E = 9 \sin(\omega t + kx + \phi)$$

Q17

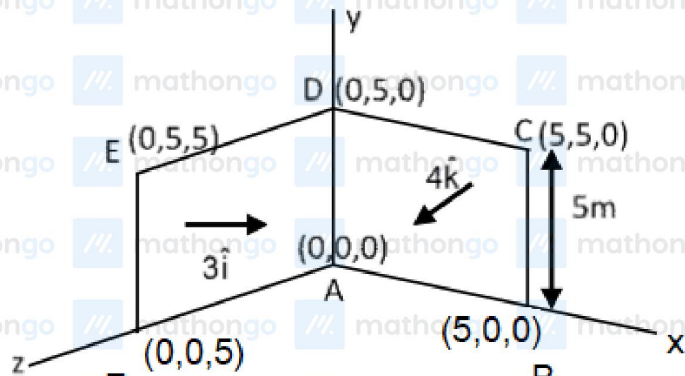
As magnetic field lines always form a closed loop, hence every magnetic field line creating magnetic flux in the inner region must be passing through the outer region. Since flux in two regions are in opposite direction,

$$\therefore \phi_i = -\phi_o$$

Q18

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$$\phi = \vec{B} \cdot \vec{A} = (3\hat{i} + 4\hat{k}) \cdot (25\hat{i} + 25\hat{k})$$

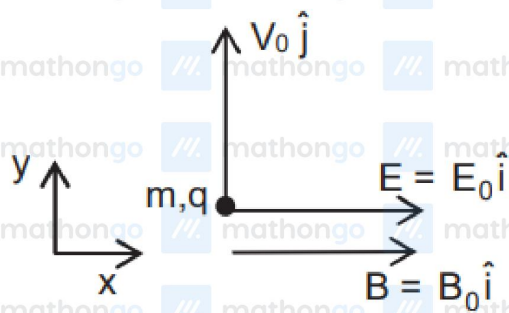
$$\phi = (3 \times 25) + (4 \times 25) = 175 \text{ weber}$$

Q19

As $\vec{v} = v_0 \hat{j}$
(magnitude of velocity does not change in y - z plane)

$$(2v_0)^2 = v_0^2 + v_x^2 ; v_x = \sqrt{3}v_0$$

$$\therefore \sqrt{3}v_0 = 0 + \frac{qE}{m}t ; t = \frac{mv_0\sqrt{3}}{qE}$$



Q20

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Physics

$$\therefore \omega = \frac{2\pi}{T} = \frac{\pi}{5}$$

$$\text{When } \omega t = \frac{\pi}{2}$$

$\therefore \phi$ will be minimum.

$\therefore e$ will be maximum

$$t = \frac{\frac{\pi}{2}}{\frac{\pi}{5}} = 2.5 \text{ sec}$$

$$\text{When } \omega t = \pi$$

$\therefore \phi$ will have maximum.

$\therefore e$ will be minimum.

$$t = \frac{\pi}{\frac{\pi}{5}} = 5 \text{ sec.}$$

Q21

Force due to electric field is in direction $-\frac{(\hat{i} + \hat{j})}{\sqrt{2}}$

because at $t = 0$, $E = -\frac{(\hat{i} + \hat{j})}{\sqrt{2}} E_0$

Force due to magnetic field is in direction $q(\vec{v} \times \vec{B})$

and $\vec{v} \parallel \hat{k}$

$\therefore$ it is parallel to $\vec{E}$

$\therefore$ net force is antiparallel to $\frac{(\hat{i} + \hat{j})}{\sqrt{2}}$.

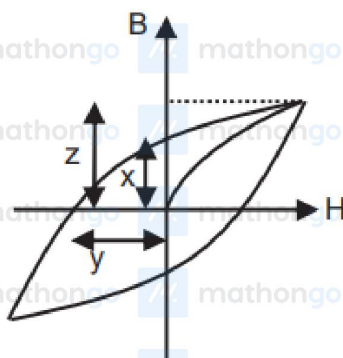
Q22

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Physics



x = retentivity

y = coercivity

z = saturation magnetization

Q23

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Physics

$$\therefore \text{K.E.} = 1.6 \times 10^{-13} = \frac{1}{2} \times 1.6 \times 10^{-27} v^2$$

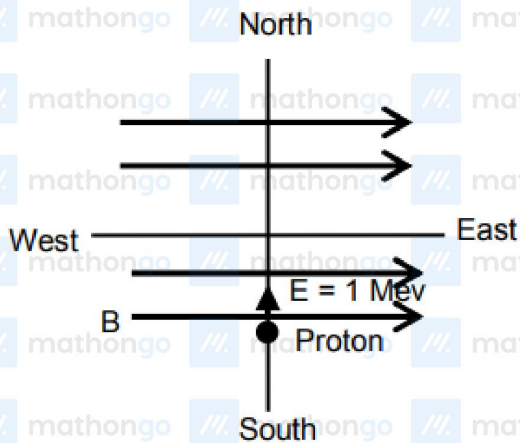
$$v = \sqrt{2} \times 10^7 \text{ m/s}$$

$$\therefore Bqv = ma$$

$$B = \frac{1.6 \times 10^{-27} \times 10^{12}}{1.6 \times 10^{-19} \times \sqrt{2} \times 10^7}$$

$$= 0.71 \times 10^{-3} \text{ T}$$

$$\text{So, } 0.71 \text{ mT}$$



Q24

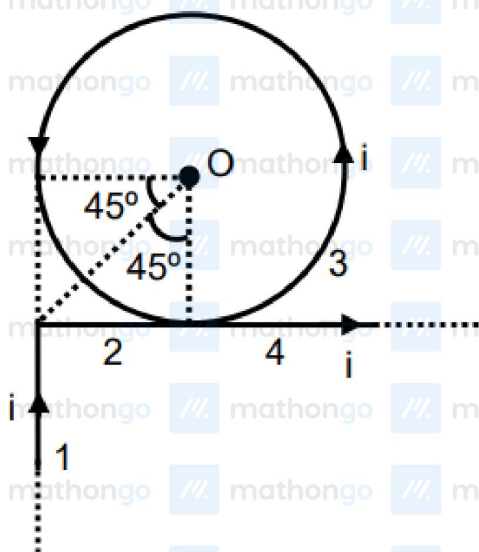
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Physics

$$\begin{aligned}
 \vec{B}_0 &= (\vec{B}_0)_1 + (\vec{B}_0)_2 + (\vec{B}_0)_3 + (\vec{B}_0)_4 \\
 &= \frac{\mu_0 i}{4\pi R} [\sin 90^\circ - \sin 45^\circ] \otimes + \frac{\mu_0 i}{2R} \odot \\
 &\quad + \frac{\mu_0 i}{4\pi R} (\sin 45^\circ + \sin 90^\circ) \odot \\
 &= \frac{-\mu_0 i}{4\pi R} \left[1 - \frac{1}{\sqrt{2}} \right] + \frac{\mu_0 i}{2R} + \frac{\mu_0 i}{4\pi R} \left[\frac{1}{\sqrt{2}} + 1 \right] \odot \\
 &= \frac{\mu_0 i}{4\pi R} \left[-1 + \frac{1}{\sqrt{2}} + 2\pi + \frac{1}{\sqrt{2}} + 1 \right] \odot \\
 &= \frac{\mu_0 i}{4\pi R} [\sqrt{2} + 2\pi] \odot \\
 &= \frac{\mu_0 i}{2\pi R} \left[\frac{1}{\sqrt{2}} + \pi \right] \odot
 \end{aligned}$$



Q25

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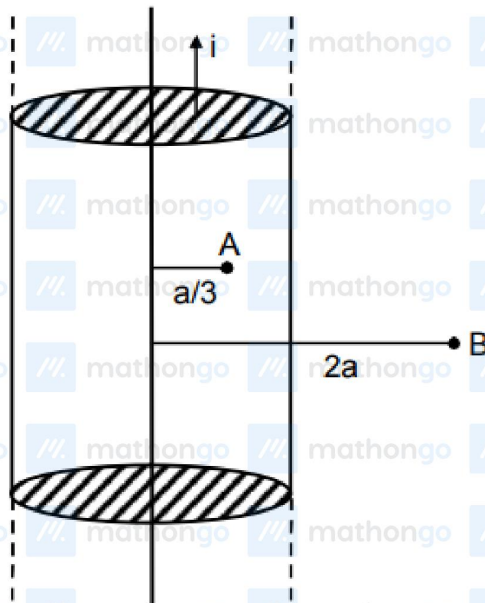
Physics

$$\frac{E}{B} = c$$

$$E = B \times c$$

$$= 15 \text{ N/C}$$

Q26



$$B_A = \frac{\mu_0 i r}{2\pi a^2} = \frac{\mu_0 i \frac{a}{3}}{2\pi a^2} = \frac{\mu_0 i}{\pi a^2} \frac{a}{6} = \frac{\mu_0 i}{6\pi a}$$

$$B_B = \frac{\mu_0 i}{2\pi (2a)}$$

$$\frac{B_A}{B_B} = \frac{4}{6} = \frac{2}{3}$$

Q27

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Physics

(A) by work energy theorem

$$W_{mag} + W_{ele} = \frac{1}{2}m(2v)^2 - \frac{1}{2}m(v)^2$$

$$0 + qE_0 2a = \frac{3}{2}mv^2$$

$$E_0 = \frac{3}{4} \frac{mv^2}{qa}$$

(B) Rate of work done at A = power of electric force

$$= qE_0 V$$

$$= \frac{3}{4} \frac{mv^3}{a}$$

(C) at Q, $\frac{dw}{dt} = 0$ for both forces

$$(D) \Delta \vec{L} = (-m2v2a\hat{k}) - (-mva\hat{k})$$

$$|\Delta \vec{L}| = 3mva$$

Q28

$$R_{\max} = \frac{R}{2} = \frac{mv_{\max}}{e\mu_0 \text{in}}$$

$$v_{\max} = \frac{Re\mu_0 \text{in}}{2m}$$

Q29

$$\tau = MB \sin \theta = I\alpha$$

$$\pi R^2 IB \theta = \frac{MR^2}{2} \alpha$$

$$\omega = \sqrt{\frac{2\pi IB}{m}} = \frac{2\pi}{T} ; T = \sqrt{\frac{2\pi m}{IB}}$$

Q30

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Physics

EM wave is in direction $\rightarrow \frac{\hat{i} + \hat{j}}{\sqrt{2}}$
Electric field is in direction $\rightarrow \hat{k}$
 $\vec{E} \times \vec{B} \rightarrow$ direction of propagation of EM wave